Experimental Comparison of EV Charging Station Ranking Algorithms

Abstract

This report evaluates the implemented EV station rankers using the repository's paired synthetic simulation. This workspace is not a business-entity matching project and contains no labeled entity pairs; matching metrics are reported as not applicable. Simulation outcomes are assumptions-based and are not field measurements.

1. Existing Pipeline and Methods

Validated station CSV loading; connector/status normalization and 10-km spatial filtering; route estimates; battery-aware charge, wait, cost, availability, reliability and rating features; then weighted scoring or a rule-based ranker. No learned classifier or training stage exists. The evaluation uses the existing paired simulation with identical generated snapshots and drivers across rankers.

2. Dataset and Configuration

Dataset	coimbatore_ev_stations_replica_demo_synthetic.csv
Stations / fields	47 / 44
Ground-truth pairs / data split	None / not applicable
Trials / drivers / seed	300 / 12 / 7
Hardware	Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz; 8 logical CPUs; 7253.9 MB RAM; Linux-7.0.0-27-generic-x86_64-with-glibc2.43
Python	3.11.16

3. Candidate Generation

Demand	Requests	Candidates	Reduction	Time (s)	Recall
low	39600	1037432	44.3%	4.7803	N/A
medium	39600	1040259	44.1%	6.4841	N/A
high	39600	1044923	43.9%	17.3004	N/A

Recall is unavailable because the data contains no labeled true matches. Reduction is measured against all station-by-request evaluations.

4. Simulated Outcomes

Mean / median / P95 values are simulated total minutes. Wait is minutes, cost is INR, and concentration is the maximum share of trial drivers assigned to one station.

Demand: low

Algorithm	Mean min	Median	P95	Wait min	>30 min	Cost INR	Max share
random	101.3	65.5	300.3	18.6	15.6%	298	17%
nearest	185.0	116.7	549.9	105.8	58.2%	289	41%
max_availability	169.7	114.5	487.7	91.4	56.5%	289	44%
fastest_rated_kw	98.8	76.1	240.4	54.8	52.1%	367	53%
cheapest	899.0	865.1	1835.9	749.3	91.1%	202	96%
naive_total_time	72.0	58.5	154.1	29.6	38.4%	364	45%
legacy_ranker (before)	83.6	70.1	188.8	36.2	44.8%	353	46%
TOPSIS (alternative)	75.2	62.8	165.1	28.8	37.9%	320	39%
generalized_cost (alternative)	70.3	56.3	150.0	24.2	32.6%	315	37%
new_ranker (after)	68.4	51.9	153.6	26.0	34.2%	343	35%
new_ranker + holds (implemented)	44.2	43.4	59.9	0.7	0.6%	339	18%

Picker/runtime resources (shared wall time and process peak RSS)

Algorithm	Calls	Picker (s)	Mean (µs)	Calls/s	Wall (s)	Peak RSS (MB)
random	3600	0.022611	6.3	159215	14.03	80.3
nearest	3600	0.022788	6.3	157979	14.03	80.3
max_availability	3600	0.046697	13.0	77093	14.03	80.3
fastest_rated_kw	3600	0.024524	6.8	146795	14.03	80.3
cheapest	3600	0.024028	6.7	149825	14.03	80.3
naive_total_time	3600	0.031283	8.7	115077	14.03	80.3
legacy_ranker (before)	3600	0.664539	184.6	5417	14.03	80.3
TOPSIS (alternative)	3600	2.757548	766.0	1306	14.03	80.3
generalized_cost (alternative)	3600	0.278845	77.5	12910	14.03	80.3
new_ranker (after)	3600	0.875470	243.2	4112	14.03	80.3
new_ranker + holds (implemented)	3600	0.943929	262.2	3814	14.03	80.3

Demand: medium

Algorithm	Mean min	Median	P95	Wait min	>30 min	Cost INR	Max share
random	117.7	75.4	350.3	33.2	28.5%	303	18%

Algorithm	Mean min	Median	P95	Wait min	>30 min	Cost INR	Max share
nearest	204.1	133.3	572.8	123.4	64.9%	292	39%
max_availability	183.8	125.2	527.6	103.6	59.8%	294	46%
fastest_rated_kw	107.2	82.6	256.8	63.2	56.2%	372	53%
cheapest	961.1	929.7	1947.0	809.5	94.2%	205	96%
naive_total_time	79.6	64.7	182.7	36.5	43.8%	365	50%
legacy_ranker (before)	110.2	84.7	264.7	57.1	53.8%	343	51%
TOPSIS (alternative)	85.8	72.7	190.0	37.5	44.7%	327	42%
generalized_cost (alternative)	71.4	56.7	153.8	24.5	32.8%	320	38%
new_ranker (after)	74.8	56.4	175.7	31.5	38.3%	347	39%
new_ranker + holds (implemented)	46.5	45.1	67.1	1.5	1.6%	343	19%

Picker/runtime resources (shared wall time and process peak RSS)

Algorithm	Calls	Picker (s)	Mean (µs)	Calls/s	Wall (s)	Peak RSS (MB)
random	3600	0.029355	8.2	122635	18.27	80.3
nearest	3600	0.031721	8.8	113490	18.27	80.3
max_availability	3600	0.065011	18.1	55375	18.27	80.3
fastest_rated_kw	3600	0.034443	9.6	104521	18.27	80.3
cheapest	3600	0.033130	9.2	108663	18.27	80.3
naive_total_time	3600	0.044508	12.4	80884	18.27	80.3
legacy_ranker (before)	3600	0.867797	241.1	4148	18.27	80.3
TOPSIS (alternative)	3600	3.461047	961.4	1040	18.27	80.3
generalized_cost (alternative)	3600	0.370719	103.0	9711	18.27	80.3
new_ranker (after)	3600	1.135852	315.5	3169	18.27	80.3
new_ranker + holds (implemented)	3600	1.170179	325.0	3076	18.27	80.3

Demand: high

Algorithm	Mean min	Median	P95	Wait min	>30 min	Cost INR	Max share
random	133.2	90.9	382.9	49.5	42.7%	302	18%
nearest	226.5	160.6	608.7	148.9	74.3%	296	40%

Algorithm	Mean min	Median	P95	Wait min	>30 min	Cost INR	Max share
max_availability	262.7	169.2	865.3	178.2	71.1%	295	59%
fastest_rated_kw	114.4	91.7	258.4	70.3	64.5%	370	52%
cheapest	995.6	952.7	1964.0	844.3	96.6%	204	96%
naive_total_time	91.3	73.8	211.0	47.1	50.1%	361	58%
legacy_ranker (before)	190.2	132.1	543.3	123.5	66.8%	320	62%
TOPSIS (alternative)	130.7	102.0	324.9	75.7	58.6%	322	57%
generalized_cost (alternative)	76.1	60.2	168.7	28.0	31.2%	321	44%
new_ranker (after)	86.3	65.4	212.7	40.9	42.1%	344	47%
new_ranker + holds (implemented)	50.6	48.4	76.6	4.0	3.7%	342	22%

Picker/runtime resources (shared wall time and process peak RSS)

Algorithm	Calls	Picker (s)	Mean (µs)	Calls/s	Wall (s)	Peak RSS (MB)
random	3600	0.080396	22.3	44778	48.53	84.0
nearest	3600	0.088661	24.6	40604	48.53	84.0
max_availability	3600	0.173927	48.3	20698	48.53	84.0
fastest_rated_kw	3600	0.097610	27.1	36881	48.53	84.0
cheapest	3600	0.096545	26.8	37288	48.53	84.0
naive_total_time	3600	0.120381	33.4	29905	48.53	84.0
legacy_ranker (before)	3600	2.277066	632.5	1581	48.53	84.0
TOPSIS (alternative)	3600	9.189062	2552.5	392	48.53	84.0
generalized_cost (alternative)	3600	0.980007	272.2	3673	48.53	84.0
new_ranker (after)	3600	2.991115	830.9	1204	48.53	84.0
new_ranker + holds (implemented)	3600	3.074874	854.1	1171	48.53	84.0

5. Accuracy, Runtime, and Interpretation

Metric	Result
Precision / recall / F1 / F0.5	N/A: no labeled entity pairs
TP / FP / FN / blocking recall	N/A: no ground truth
Training time	N/A: no training stage
Runtime significance	Not tested; picker timing is descriptive
Memory	Process-level peak RSS; not per algorithm

Shared preprocessing (station loading plus route-table construction) took 0.0214 seconds. Feature extraction is performed within some picker calls and is not isolated; picker timing includes that work. Wall time covers all algorithms and the complete simulation for one demand level. Because the outcome model uses the current battery-aware charging estimator and all data are synthetic, these values do not establish real-world superiority. Reservation-aware selection changes system behavior by coordinating reservations and is not a pure ranker comparison.

Measured scenario winners

At low demand, new_ranker (after) had the lowest mean simulated total time among pure rankers (68.35 min); random had the lowest mean picker time (6.3 µs/call). Reservation-aware selection reduced mean wait from 25.99 to 0.67 min (97.4%) in this simulator, but adds coordination and is not a pure ranker.

At medium demand, generalized_cost (alternative) had the lowest mean simulated total time among pure rankers (71.41 min); random had the lowest mean picker time (8.2 µs/call). Reservation-aware selection reduced mean wait from 31.47 to 1.51 min (95.2%) in this simulator, but adds coordination and is not a pure ranker.

At high demand, generalized_cost (alternative) had the lowest mean simulated total time among pure rankers (76.12 min); random had the lowest mean picker time (22.3 µs/call). Reservation-aware selection reduced mean wait from 40.92 to 3.96 min (90.3%) in this simulator, but adds coordination and is not a pure ranker.

Limitations

The dataset has 47 synthetic station records and approximate coordinates. There are no duplicate labels, entity splits, external traffic observations, or real charge sessions. Runtime is a single-process measurement on one machine. No statistical significance is claimed for compute timings.